

# StudyPilot FullRun Demo

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## Summary

- The first law states that an object at rest stays at rest and an object in motion stays in motion unless acted on by a net external force.

- This

## Key Terms

- Motion

- Obj

## Q&A History

**\*\*Q:** What is Newton's second law?\*\*

A: Today we are going to talk about Newton's laws of motion. The first law states that an object at rest stays at rest and an object in motion stays in motion unless acted on by a net external force. This is also called the law of inertia. Inertia is the tendency of an object to resist changes in its state of motion. The second law relates force, mass, and acceleration with the equation  $F$  equals  $m$  times  $a$ . This means the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. The third law states that for every action there is an equal and opposite reaction. A classic example is

## From Transcript

Today we are going to talk about Newton's laws of motion. The first law states that an object at rest stays at rest and an object in motion stays in motion unless acted on by a net external force. This is also called the law of inertia. Inertia is the tendency of an object to resist changes in its state of motion. The second law relates force, mass, and acceleration with the equation  $F = ma$ . This means the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. The third law states that for every action there is an equal and opposite reaction. A classic example is a rocket engine, where gas is expelled backward and the rocket is pushed forward. Let's do a quick example problem. Suppose a car of mass 1000 kilograms accelerates at 2 meters per second squared. Using  $F = ma$ , the net force required is 2000 newtons. Momentum is another important concept, defined as mass times velocity. In the absence of external forces, the total momentum of a closed system is conserved. This is called the conservation of momentum, and it's very useful for analyzing collisions. There are two types of collisions we study: elastic collisions, where kinetic energy is conserved, and inelastic collisions, where it is not. In a perfectly inelastic collision, the two objects stick together after impact. For the exam, make sure you can derive the equations for momentum conservation in both elastic and inelastic collisions. Also review free body diagrams, since most problems on the midterm will ask you to identify all forces acting on an object. Next class we will cover circular motion and centripetal force, so please read chapter five before then.